

Final Test by Statistic

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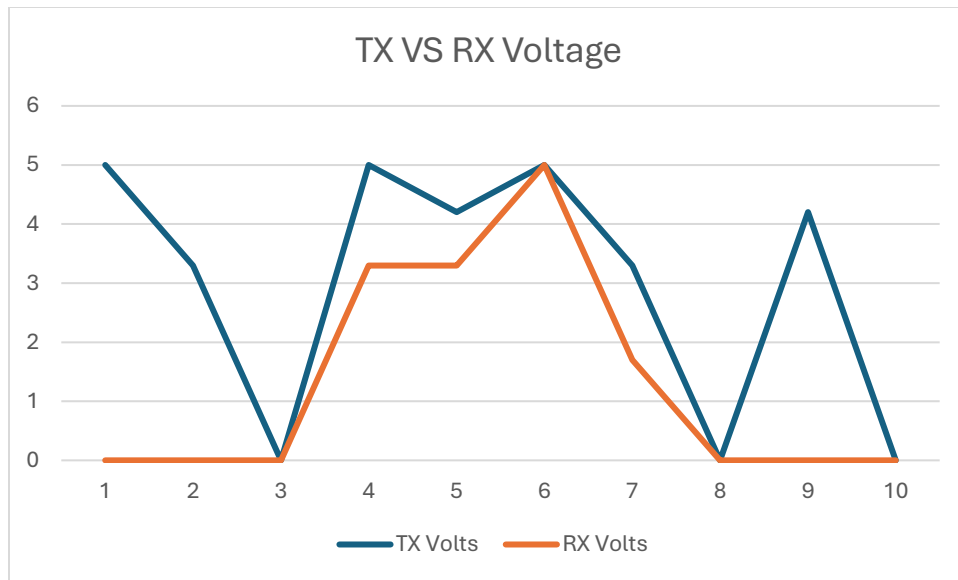
Introduction

The final test by statistics will measure 1's and 0's and voltage based on the keying of the transmitter of the flex sensors and the signal transmitted to the receiver module and is decoded.

Measurements/Calculations

Trial	TX volt	RX Volt
1	5V	0V
2	3.3V	0V
3	0V	0V
4	5V	3.3V
5	4.2V	3.3V
6	5V	5V
7	3.3V	1.7V
8	0V	0V
9	4.2V	0V
10	0V	0V

Mean TX	3.25V
Range TX	5V
Std Dev TX	2.07V
Mean RX	1.33V
Range RX	5V
Std Dev RX	1.789V



Analysis and Conclusion

Analysis of the TX and RX voltage data reveals that there is a decoding reliability issue within the receiver module. Even though the transmitter was able to consistently output the expected voltage levels in response to the flex sensor inputs, the receiver module failed to mirror the values within the same consistency. This could point to bit drop errors during signal decoding, which could be caused by timing mismatch, protocol not engaging properly or insufficient signal strength.